

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization and over 4 years of experience as a Software Developer and Data Engineer, adept at building scalable microservices and distributed data pipelines. Possesses a Master of Science in Computer Science with demonstrated expertise in Java, Python, Spark, SQL, AWS, and Azure, complemented by CI/ CD automation and graduate-level research in large-scale data platforms and privacy-preserving AI. Seeking a Software Engineer - Backend role to leverage full-stack development skills and contribute to robust backend solutions.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance REST and SOAP microservices using Java 8/11, Spring Boot, and Azure Functions, managing over 50,000 daily API requests with 99.99% SLA uptime.
- Automated CI/ CD pipelines with Jenkins, Maven, and Azure DevOps, enabling continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x and enhancing monitoring, reducing issue resolution time by 20%.
- Led a significant EDX platform upgrade, streamlining application architecture and dependency management to achieve 50% faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions utilizing Git, SVN, and comprehensive JUnit testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed robust Java and EJB-based enterprise applications and RESTful web services for automotive clients.
- Optimized system performance by 90% through advanced SQL query tuning, database enhancements, and legacy Java code modernization.
- Implemented CI/ CD pipelines leveraging Jenkins and GitLab, integrating automated testing and quality gates for efficient deployments.
- Mentored junior developers through rigorous code reviews, debugging assistance, and promoting clean coding practices.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high- availability enterprise systems, ensuring over 99% uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10% and reinforcing data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99% code coverage for critical modules.
- Supported release planning and sprint execution using JIRA within Agile development frameworks.

PROJECTS

NYU Taxi Data Analysis Platform January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real time.
- Built an interactive Flask web application for data exploration with dynamic filtering and visualization capabilities.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments across Linux environments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40%.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight.

Privacy-Preserving AI Email Client (Capstone Project) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, exploring caching strategies for performance.

SKILLS

- Programming Languages: Java, Python, SQL, Linux.
- Frameworks and Libraries: Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, JUnit, Ansible.
- Web Services: REST API, SOAP API, Fullstack development.
- Cloud Platforms: AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), Google Cloud Platform.
- Databases: PostgreSQL, Oracle Database, IBM DB2, RDS.
- Tools and Technologies: Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, Terraform, Tomcat, Github.
- Concepts: ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/ CD, Microservices, Large- scale data platforms, Infrastructure automation, Privacy- preserving LLM- based applications,

Authentication, Account Management.

- Data Analysis and Visualization: Tableau, QuickSight.